

Date : 20/8/2026

Name : K. Rishika

Course : CSA0303

Regno : 192525114

CO4 AT-1

Comparative Analysis :

1. Compare the Hashing and Binary Tree based on Time Complexity and Methods.
2. Compare the Time complexity of all the notations
i). Worst Case ii). Best Case iii). Average Case
in bubble, insertion, selection, merge, Quick, Heap Sorting.

Answers:

Hashing	Binary Tree
* Search : Average $O(1)$, Worst $O(n)$.	* Search : Average $O(\log n)$, Worst $O(n)$.
* Insertion : Average $O(1)$, Worst $O(n)$	* Insertion : Average $O(\log n)$, Worst $O(n)$,
* Deletion : Average $O(1)$, Worst $O(n)$.	* Deletion : Average $O(\log n)$, Worst $O(n)$.
* Use hash Function	* Uses comparison of keys.
* Data is not ordered	* Data is ordered
* Faster for direct Searching	* Useful for sorted and range-based data.
* Uses hash table.	* Uses tree structure.

2 Time Complexity of Sorting Algorithms:

Sorting Algorithm	Best Case	Average Case	Worst Case
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$
Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$

